

Mean Absolute Deviation

Lesson 8-3

Name: _____

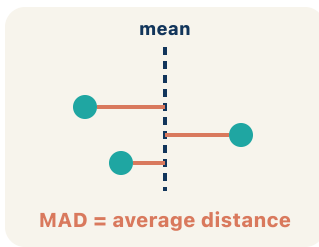
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

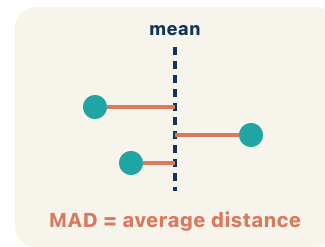
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Data: 8, 10, 12. Mean = 10. Distances from mean:
2, 0, 2. $MAD = (2+0+2) \div 3 = 1.33$

Mean Absolute Deviation

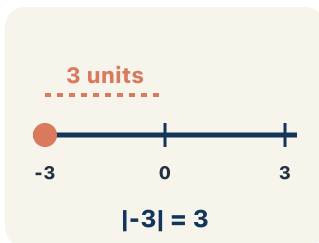
Write the definition:



If mean = 20 and value = 17, deviation = $17 - 20 = -3$ (3 below the mean)

Deviation

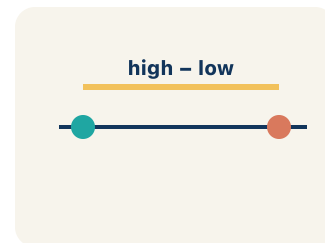
Write the definition:



$|-3| = 3$ and $|3| = 3$ — both are 3 units from zero

Absolute Value

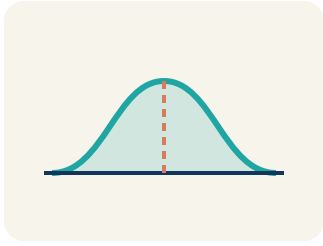
Write the definition:



Low spread: 8, 9, 10, 11 (close together). High spread: 2, 9, 10, 25 (far apart)

Spread

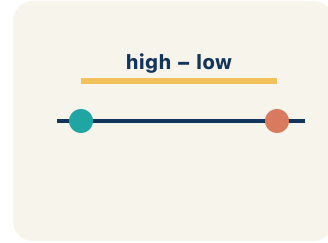
Write the definition:



A set clustered tightly around the mean has low MAD; a set spread far from the mean has high MAD

Data distribution

Write the definition:



Low variability (MAD = 1): very consistent. High variability (MAD = 8): very spread out

Variability

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the mean absolute deviation (MAD) to describe how spread out data is.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean Absolute Deviation

Deviation

Absolute Value

Spread

Data distribution

Variability

1 The average distance of each value from the mean is the _____.

2 How far a single value is from the mean is its _____.

3 The distance of a number from zero, always positive, is its _____.

4 How far apart the data values are from each other is the _____.

5 The way data values are spread out is the _____.

6 How much the data values differ is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

- A. 4
- B. -4
- C. 26
- D. 11

Step 1 Deviation = $11 - 15 = -4$.

Step 2 Absolute deviation = $|-4| = 4$.

 **Answer:** A. 4

 We do — together

Solve this one with your class using the same steps.

Problem: A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

- A. 3
- B. 5
- C. 15
- D. 1

Step 1 _____

Step 2 _____

Answer: _____

**You do — your turn**

Now try one on your own. Show every step.

Problem: The mean of a data set is 20. A value is 26. What is the absolute deviation?

- A. 6
- B. -6
- C. 46
- D. 26

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A guard scored 2, 4, 6, 8 points in four scrimmages, so the mean is 5 points. The distances from the mean are 3, 1, 1, 3. What is the mean absolute deviation (MAD)?

- A. 1
- B. 2
- C. 5
- D. 8

Show your work:

2. Two shooters average 11 points a game. Shooter A's MAD is 2 points and Shooter B's MAD is 6 points. Which shooter is MORE consistent?

- A. Shooter A, because a smaller MAD means scores stay closer to the average
- B. Shooter B, because a bigger MAD means a bigger average
- C. They are equally consistent because they have the same mean
- D. You cannot tell consistency from MAD

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two basketball teams both average 75 points per game. Team A's last 5 scores: 73, 76, 74, 77, 75. Team B's last 5 scores: 60, 90, 65, 85, 75. Calculate the MAD for each team and explain which team a coach would prefer if they need predictable scoring.

Sentence starter: Team A's MAD = _____. Team B's MAD = _____. Team A is _____ because _____. A coach would prefer Team _____ for predictable scoring because _____.

Show your work:

Reflect — Exit Ticket

Data set: 4, 8, 6, 10, 2. The mean is 6. What is the MAD?

- A. 2.4
- B. 6
- C. 0
- D. 12

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **We Try (notes):** A. 3 — *Sum of absolute deviations = $3 + 1 + 5 + 2 + 4 = 15$. $MAD = 15 \div 5 = 3$.*
8. **You Try (notes):** A. 6 — *Deviation = $26 - 20 = 6$. Absolute deviation = $|6| = 6$.*
9. **Try It 1:** B. 2 — *Add the distances: $3 + 1 + 1 + 3 = 8$. Divide by the 4 games: $8 \div 4 = 2$. The MAD is 2 points.*
10. **Try It 2:** A. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*
11. **Exit Ticket:** A. 2.4 — *Deviations: $-2, 2, 0, 4, -4$. Absolute deviations: $2, 2, 0, 4, 4$. Sum = 12. $MAD = 12 \div 5 = 2.4$.*